When we examine the Coma cluster, as Zwicky did during the 1930s, we find that its member galaxies are all moving more rapidly than the escape velocity for the cluster. The cluster should swiftly fly apart, leaving barely a trace of its beehive existence after just a few hundred million years had passed. But the cluster is more than ten billion years old, which is nearly as old as the universe itself. And so was born what remains the longest-standing unsolved mystery in astrophysics.

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Across the decades that followed Zwicky's work, other galaxy clusters revealed the same problem, so Coma could not be blamed for being peculiar. Then what or who should we blame? Newton? I wouldn't. Not just yet. His theories had been examined for 250 years and passed all tests. Einstein? No. The formidable gravity of galaxy clusters is still not high enough to require the full hammer of Einstein's general theory of relativity, just two decades old when Zwicky did his research. Perhaps the "missing mass" needed to bind the Coma cluster's galaxies does exist, but in some unknown, invisible form. Today, we've settled on the moniker "dark matter," which makes no assertion that anything is missing, yet nonetheless implies that some new kind of matter must exist, waiting to be discovered.

Just as astrophysicists had come to accept dark matter in galaxy clusters as a mysterious thing, the problem reared its invisible head once again. In 1976, the late Vera Rubin, an astrophysicist at the Carnegie Institution of Washington, discovered a similar mass anomaly within spiral galaxies themselves. Studying the speeds at which stars orbit their galaxy centers, Rubin first found what she expected: within the visible disk of each galaxy, the stars farther from the center move at greater speeds than stars close in. The farther stars have more matter (stars and gas) between themselves and the galaxy center, enabling their higher orbital speeds. Beyond the galaxy's luminous disk, however, one can still find some isolated gas clouds and a few bright stars. Using these objects as tracers of the gravity field exterior to the most luminous parts of the galaxy, where no more visible matter adds to the total, Rubin discovered that their orbital speeds, which should now be falling with increasing distance out there in Nowheresville, in fact remained high.

These largely empty volumes of space—the far-rural regions of each galaxy—contain too little visible matter to explain the anomalously high orbital speeds of the tracers. Rubin correctly reasoned that some form of